

Predicting Odor Descriptors from Molecular Structure

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1 Data and Features

1.1 Dataset

The dataset merges two public odor-labeled molecule collections, GoodScents and Leffingwell, deduplicated by canonical SMILES. Each molecule is annotated with one or more of 138 odor descriptors, which are clustered into 12 coarse metacategories.

Property	Value
Molecules	4,981
Source	GoodScents + Leffingwell
Fine-grained labels	138
Metacategories	12
Label type	Multi-label
Metacategory prevalence (min / median / max)	10.3% / 22.9% / 45.8%
Input representation	SMILES string

Table 1: Dataset summary.

1.2 Feature Construction

Hypothesis

Three complementary feature families, computed directly from the SMILES string, are enough to describe a molecule for odor prediction: substructure keys, local circular fingerprints, and continuous physicochemical descriptors.

Implementation

Each SMILES is canonicalized with RDKit, then three feature blocks are computed:

- **MACCS** : 166 binary bits, each flagging whether a specific, predefined substructure is present (`MACCSkeys.GenMACCSKeys`).
- **Morgan** : 512 binary bits encoding circular atomic neighborhoods, radius = 2 (`rdFingerprintGenerator.GetMorganGenerator(radius=2, fpSize=512)`).
For each atom, the algorithm looks at its local neighborhood (up to 2 bonds away), encodes that neighborhood as a substructure, and hashes it into one of 512 bit positions. This is done for every atom, so the final fingerprint is a bit vector marking which circular substructures appear in the molecule.
- **Mordred** : 327 continuous 2D descriptors (topological indices, autocorrelations, VSA descriptors, ring counts). These describe the molecule as a whole (e.g. molecular weight, LogP).

Results : This gives $166 + 512 + 327 = 1,005$ structural features per molecule. Five molecules with missing descriptor values were dropped, leaving **4,976 molecules**.

Preprocessing Pipeline

Three filtering steps are applied, each computed on the training split only and then applied identically to the test split to avoid leakage:

- **Zero-variance removal.** Columns that are identical across every molecule carry no information and are dropped independently. This removes 6 of the 678 fingerprint bits (MACCS + Morgan); Mordred loses none.
- **Standardization.** The Mordred block is scaled with `StandardScaler`, fit on train only. This is necessary because Mordred descriptors has heterogeneous physical scales, while the binary fingerprint need no scaling.
- **Correlation filter.** Mordred features with pairwise Pearson $|r| > 0.95$ are considered redundant. For each correlated pair, one feature is dropped; this removes 6 of the 327 Mordred descriptors, leaving 321.

Result : After filtering, the fixed structural input is 672 (fingerprints) + 321 (Mordred) = 993 features per molecule.

2 Train / Validation / Test Split

2.1 Hypothesis

A naive random split will bias evaluation on a severely imbalanced multi-label dataset, because random sampling cannot prevent rare labels from being absent in a split; only multi-label stratification keeps every split representative.

2.2 Implementation

Splits use `MultilabelStratifiedShuffleSplit` (the `iterative-stratification` package), stratified on the 138-dimensional fine-label matrix Y_1 , so label prevalence is preserved across splits. Two split regimes appear in this report :

- **Classical baselines (BR variants):** 80% train / 20% test, no validation set because not needed.
- **ChemBERTa and HMCN-F:** 60% train / 20% validation / 20% test, the validation set is cut out from training to allow early stopping and threshold calibration.

3 Transformer Representations (ChemBERTa)

Fingerprints encode presence/absence of a fixed vocabulary of substructures and know nothing about smell. ChemBERTa is a transformer pretrained on 77M SMILES from ZINC by masked-language-modeling, predicting masked tokens of a SMILES from context. The bet is that this is a dense, continuous representation capturing richer structural relationships than fixed vocabulary fingerprints, and that some of this transfers to odor.

- **H1:** Frozen ChemBERTa embeddings contain at least as much odor information as MACCS + Morgan + Mordred, measured by macro PR-AUC.

- **H2:** If frozen embeddings fall short, fine-tuning will close or exceed the gap to the baseline.

3.1 Shared Setup

All four experiments share dataset, splits, loss, and evaluation.

- **Model:** seyonec/ChemBERTa-zinc-base-v1 : 6 transformer layers, 768-dim hidden state.
- **Split:** 60/20/20 train/val/test, multilabel-stratified. (Baseline used 80/20 with no val set, so ChemBERTa trains on less data)
- **Loss:** Asymmetric Loss (ASL). Plain BCE is dominated by easy negatives under imbalance; ASL down weights easy negatives and focuses gradient on rare positives:

$$\mathcal{L} = -y(1-p)^{\gamma^+} \log p - (1-y)p_m^{\gamma^-} \log(1-p_m), \quad p_m = \max(p-m, 0). \quad (1)$$

- **Threshold calibration:** after training, search a per-label decision threshold on validation over [0.1, 0.9] of step 0.02 maximizing per-label F1. PR-AUC and ROC-AUC are threshold-independent and unaffected.
- **Primary metric:** macro PR-AUC.

3.2 Experiment 1 : Frozen ChemBERTa + Linear Head

How It Was Implemented

The backbone is frozen; its weights never change. Each SMILES is passed through once to extract a fixed 768-dim [CLS] embedding, cached, and a single linear layer is trained on top:

$$h = \text{ChemBERTa}(\text{SMILES})_{[\text{CLS}]} \in \mathbb{R}^{768}, \quad \hat{y} = \sigma(Wh + b), \quad W \in \mathbb{R}^{12 \times 768}. \quad (2)$$

Config: optimizer Adam, LR = 10^{-3} ; ASL ($\gamma^+=0$, $\gamma^-=4$, $m=0.05$); 300 epochs.

Results

Metric	Exp 1 (Linear)	BR+RF
macro PR-AUC	0.548	0.624
Macro ROC-AUC	0.761	0.820
Macro F1	0.545	0.567

Label	BR+RF	Frozen + Linear
fruity	0.820	0.748
chemical	0.717	0.679
green	0.712	0.676
floral	0.709	0.595
gourmand	0.638	0.578
sweet	0.610	0.607
woody	0.592	0.517
spicy	0.562	0.462
earthy	0.568	0.537
powdery_amber	0.556	0.412
animal_musk	0.527	0.394
citrus	0.470	0.377
Macro avg.	0.624	0.548

Table 2: Per-label macro PR-AUC

Analysis

The model converged around epoch 35 (val macro PR-AUC ≈ 0.529) and plateaued for the remaining 265 epochs. The plateau is the key observation: more epochs do nothing, so the linear head has already extracted everything a linear map can from the fixed embedding. Result: 0.076 below baseline.

3.3 Experiment 2 : Frozen ChemBERTa + Shallow MLP Head

How It Was Implemented

Identical to Experiment 1 , but the linear head is replaced by a two-layer MLP, to test whether the linear head was under-fitting a representation that actually holds non-linear structure:

$$\hat{y} = W_2 \text{Dropout}(\text{ReLU}(W_1 h + b_1)) + b_2, \quad W_1 \in \mathbb{R}^{256 \times 768}, W_2 \in \mathbb{R}^{12 \times 256}, \text{ dropout} = 0.3. \quad (3)$$

The 256-unit hidden layer and 0.3 dropout are a moderate, conventional capacity, enough to model interactions among the 768 dims, still small enough not to overfit, since only the head trains.

Results

Metric	Exp 1 (Linear)	Exp 2 (MLP)	BR+RF
macro PR-AUC	0.548	0.561	0.624
Macro ROC-AUC	0.761	0.775	0.820
Macro F1	0.545	0.552	0.567

Analysis

The MLP gains +0.013 macro PR-AUC over the linear head . But after epoch 65 the training loss kept falling (0.134 $\rightarrow$ 0.096) while val macro PR-AUC stopped improving and drifted down , overfitting, the MLP fitting training noise rather than finding new signal.

Conclusion

The bottleneck is not the head’s capacity , it is the information content of the frozen embedding , so a more complex head will not help.

3.4 Experiment 3 :Partial Fine-Tuning (Top-2 Layers Unfrozen)

Implementation

The backbone is partly unfrozen: embedding layer and lower transformers stay frozen; the **top 2 of 6** transformer layers train jointly with the MLP head. Two learning rates protect the pretrained weights:

Parameter	Value
Unfrozen layers	Top 2 of 6
Backbone LR	1×10^{-5}
Head LR	1×10^{-4}
LR schedule	Linear warmup , cosine decay
ASL	$\gamma^+=0$, $\gamma^-=4$, $m=0.05$
Label smoothing	$\varepsilon = 0.05$ on ASL targets
Early stopping	on validation macro PR-AUC, patience 10

Extra regularization specific to fine-tuning: **SMILES enumeration** , 20 random valid SMILES per training molecule (e.g. CCO, OCC, C(O)C all mean ethanol), re-sampled each epoch, forcing the model to learn structure rather than memorize one string.

Results

Metric	Exp 2 (Frozen MLP)	Exp 3 (Fine-tuned)	BR+RF
macro PR-AUC	0.561	0.559	0.624
Macro ROC-AUC	0.775	0.778	0.820
Macro F1	0.552	0.552	0.567

Label	BR+RF	Frozen + Linear	Frozen + MLP	Fine-tuned
fruity	0.820	0.748	0.757	0.734
chemical	0.717	0.679	0.699	0.674
green	0.712	0.676	0.693	0.703
floral	0.709	0.595	0.626	0.636
gourmand	0.638	0.578	0.597	0.622
sweet	0.610	0.607	0.603	0.593
woody	0.592	0.517	0.526	0.549
earthy	0.568	0.537	0.543	0.544
spicy	0.562	0.462	0.451	0.453
powdery_amber	0.556	0.412	0.412	0.396
animal_musk	0.527	0.394	0.426	0.435
citrus	0.470	0.377	0.401	0.362
Macro avg.	0.624	0.548	0.561	0.559

Analysis

Even letting the transformer’s weights adapt to the odor data, with augmentation and proper regularization, produces no gain over the frozen MLP (0.559 vs 0.561). The gap to BR+RF (0.065) is essentially unchanged from Experiment 2 (0.063).

Conclusion

Adaptation of the representation does not close the gap.

3.5 Experiment 4 : Concatenation of Fingerprints + Frozen ChemBERTa

Implementation

Instead of replacing fingerprints, I combined them. Concatenate the frozen 768-dim [CLS] embedding with the full 1,005-dim fingerprint+Mordred vector and feed the result to the same BR+RF pipeline as the baseline:

$$x_{\text{combined}} = [x_{\text{fingerprints}}(1005); x_{\text{ChemBERTa}}(768)] \in \mathbb{R}^{1773} \longrightarrow \text{BinaryRelevance}(\text{RandomForest}, 300 \text{ trees}). \quad (4)$$

The question: does ChemBERTa carry any complementary signal that improves the baseline when added on top, even if it cannot replace fingerprints?

Results

Metric	Fingerprints only	Fingerprints + ChemBERTa	Change
macro PR-AUC	0.624	0.624	+0.000
Macro ROC-AUC	0.820	0.815	−0.005
Macro F1	0.567	0.597	+0.030

Analysis

macro PR-AUC is flat (difference ≈ 0.0001) and ROC-AUC is flat, which means ChemBERTa added no new discriminative ranking signal on top of fingerprints..

Conclusion

No complementary signal on the primary metric. .

3.6 ChemBERTa Summary and Conclusion

Configuration	macro PR-AUC	ROC-AUC	F1
BR+RF, fingerprints only	0.624	0.820	0.567
Frozen ChemBERTa + Linear head	0.548	0.761	0.545
Frozen ChemBERTa + MLP head	0.561	0.775	0.552
Fine-tuned ChemBERTa (top-2) + MLP	0.559	0.778	0.552
Fingerprints + Frozen ChemBERTa $\rightarrow$ BR+RF	0.624	0.815	0.597